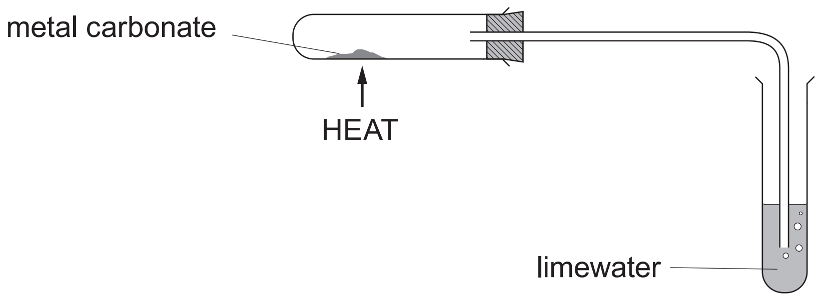
Examiner only

6. (a) Rhian investigated the decomposition of three different metal carbonates.

She measured the time taken for limewater to turn milky using the following apparatus.



Her results are shown in the table.

Metal carbonate	Time taken for limewater to turn milky (s)
copper(II) carbonate	18
zinc carbonate	27
lead carbonate	11

(i) Place the carbonates in order of stability. [1]

Most stable ....

....

Least stable ....

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(ii) If sodium carbonate was used in the investigation the limewater would not turn milky however long it was heated.

Tick (✓) the reason why the limewater would not turn milky. [1]

Sodium carbonate only decomposes a small amount on heating	☐
Sodium carbonate is very unstable	☐
Sodium carbonate does not decompose on heating	☐
Sodium carbonate decomposes too quickly	☐

(iii) On heating copper(II) carbonate, Rhian expected to make 5.0 g of copper(II) oxide. She actually made 3.5 g.

Use the formula below to calculate the percentage yield of copper(II) oxide in her experiment. [2]

$$\text{percentage yield} = \frac{\text{actual mass}}{\text{expected mass}} \times 100$$

Percentage yield = ... %

(iv) One of the ions present in copper(II) carbonate is CO_3^{2-}. [1]

Give the formula of the other ion present.

....

(b) Rhian carried out a flame test to show that sodium carbonate contains sodium ions.

Give the colour of the flame seen. [1]

....

6

